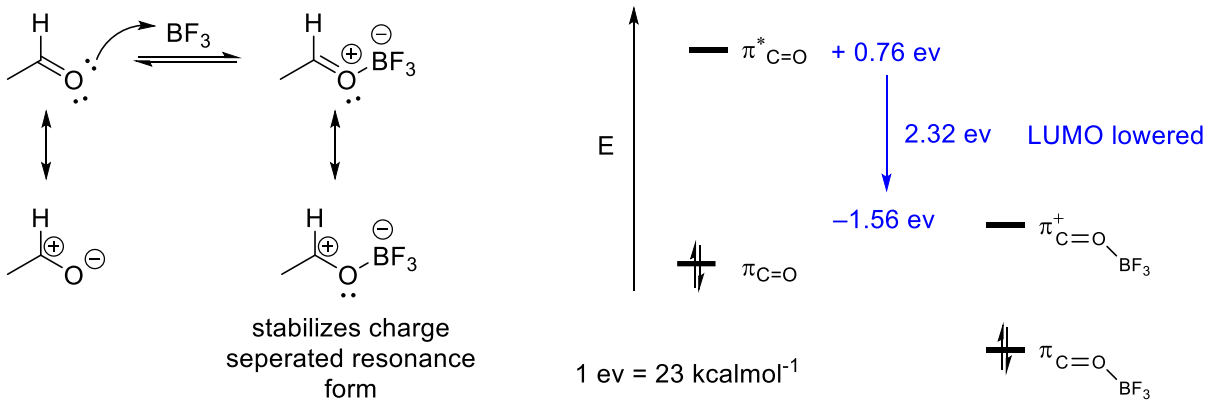
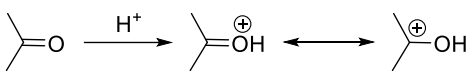
Class 8

Reactivity of carbonyl compounds can be increased using a LUMO lowering catalyst

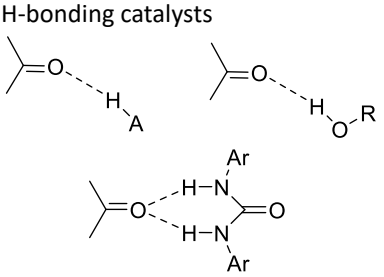


LUMO lowering catalysts

- Lewis acid catalysts (BF_3, $AlCl_3$, $ZnCl_2$.....)
- Bronsted acid catalyst

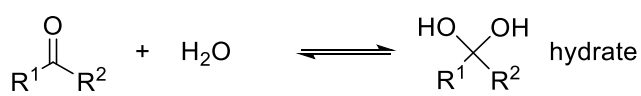


- H-bonding catalysts

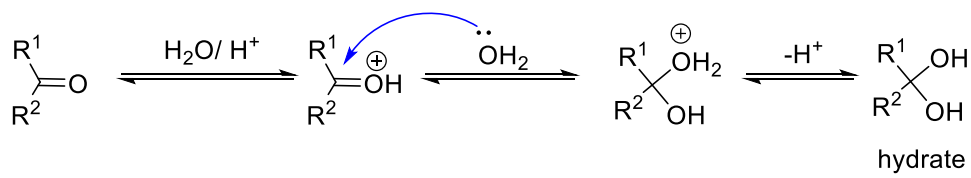


Reactions of Oxygen Nucleophiles

Hydration reaction:

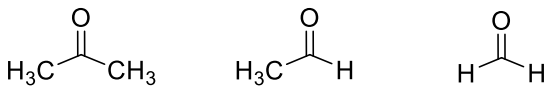


Mechanism:



- The reaction is reversible
- Position of the equilibrium is important
- Depends upon relative stability of ketone vs hydrate

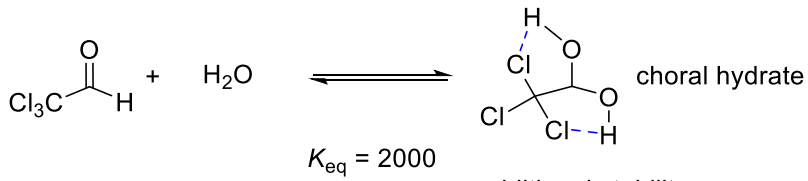
Examples:



Keq	0.002	1	2300

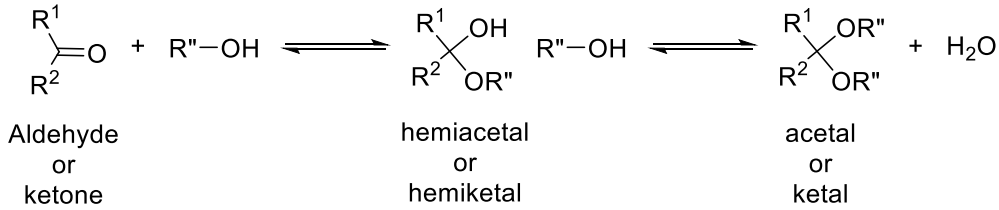
amount of hydrate increasing →

- **Sterics:** hydrate is more crowded, less bulky substrate leads to more hydrate
- **Electronics:** $\sigma_{C\text{-}H} \rightarrow \sigma^*_{C\text{-}O}$ less → more hydrate

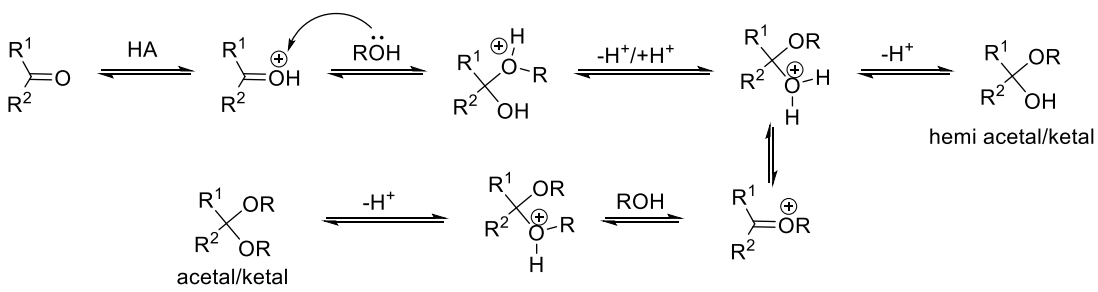


(Steric is not good argument, electronics: 1. Inductive effect 2. MO 3. H- bonding)

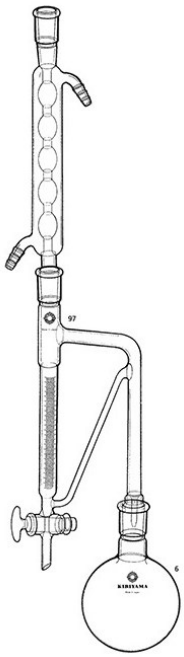
Reactions with alcohol:



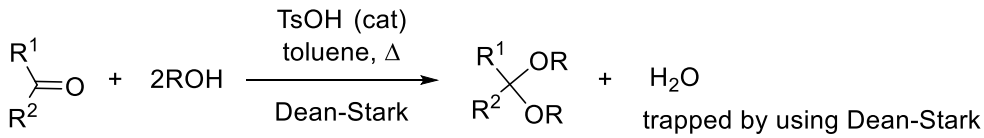
Mechanism: (acid catalysis)



- The reaction is in equilibrium. Removal of water is required to shift the equilibrium.
- Water is usually removed azeotropically by using a Dean – Stark apparatus.

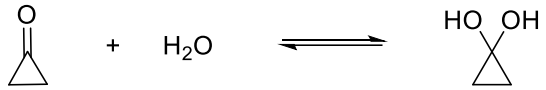


General Procedure:

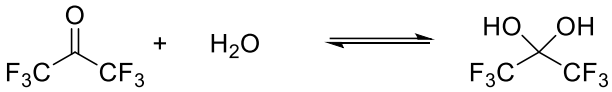


Questions:

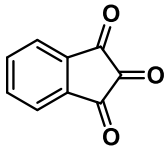
1. Predict the position of the following equilibrium.

(a) 

(c)

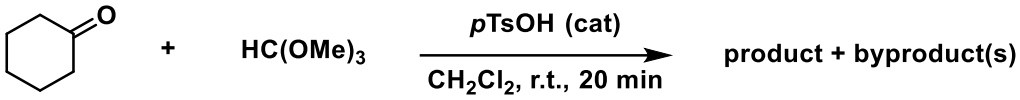
(b) 

2. The triketone shown below is called Ninhydrin and exists in aqueous solution as a monohydrate. Which of the three keto groups gets hydrated? Explain.

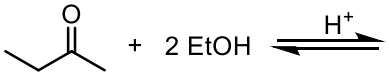


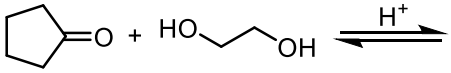
Ninhydrin

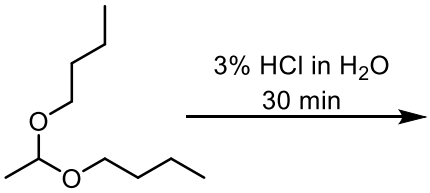
3. Write the product and byproduct for the following reaction with an appropriate arrow pushing mechanism.

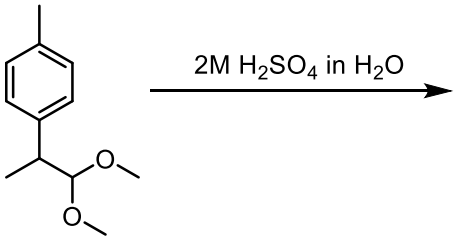


4. Write the product for the following reaction with an appropriate arrow pushing mechanism.

(a) 

(b) 

(c) 

(d) 

5. Explain the concept of LUMO lowering catalysis with an appropriate example.